Wassim Kabalan

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Profile

I am a Research Associate at The Alan Turing Institute, where I work on AI for physics. During my PhD in computational cosmology I built differentiable simulation and inference software in **JAX**, from the statistical method down to the distributed GPU code, and several of these libraries are now used by other groups. Before the PhD I spent five years writing production **C++** at Dassault Systèmes inside a multi-million-line codebase.

Education & Professional Experience

Oct 2026 – present **Research Associate**, *The Alan Turing Institute*, London
AI for Physics team (Fundamental Research), with Jason McEwen. Physics-informed emulators and rigorous Bayesian uncertainty quantification.

Dec 2023 – Oct 2026 **PhD in Computational Cosmology**, *Université Paris Cité — APC (CNRS/IN2P3)*, Paris
Thesis: Differentiable Field-Level Cosmology: CMB, Weak Lensing, and GPU-Accelerated Inference.
Advisors: Eric Aubourg, Alexandre Boucaud, Josquin Errard, François Lanusse.

- *End-to-end differentiable simulation (thesis core).* Built one of the largest differentiable N-body simulations to date — a single pipeline running from the initial conditions of the universe to the projected matter distribution on the sky, then inverted to recover the cosmological parameters that govern our universe. Fully GPU-accelerated and distributed with **JAX** sharding I ran it on **512 H100 GPUs** (∼**70 billion particles**), with MCMC over a $\sim 10^9$-dimensional posterior. Engineered the entire stack, from the gravity solver to the spherical projection and the inference, with a focus on **scalability**, **differentiability**, and **memory efficiency**.
- *Cosmological signal decontamination (Simons Observatory).* Separating the faint primordial signal from bright astrophysical foregrounds is the bottleneck for next-generation CMB satellites. I implemented a novel adaptive, gradient-based minimizer (`ADABK`) purpose-built for this regime — extremely low signal-to-noise, hard physical bounds, and spatially-varying foregrounds — reaching better solutions in far fewer iterations than standard optimizers. First author of the resulting paper (arXiv:2604.08463).

Oct 2019 – Oct 2023 **Software Infrastructure Engineer (3DEXPERIENCE & CATIA)**, *Dassault Systèmes*, Vélizy-Villacoublay

- Built and maintained core infrastructure inside a **multi-million-line modern C++ (17/20)** codebase on **Linux**, depended on by hundreds of engineers across teams (**CATIA**, **SOLIDWORKS**, **3DEXPERIENCE**).
- Designed data-interoperability and conversion services across CAD products, and the CATIA cache infrastructure — including a *multi-session distributed cache* — with a focus on **parallel and distributed computing**.
- Owned the **CI/CD** pipelines and the test matrix for the conversion-service infrastructure.

Jan 2019 – Oct 2019 **Data Acquisition Engineer (Renault)**, *SERMA*, Guyancourt

- Built tooling for high-volume vehicle sensor logs: a **C** decoder for ECU binary data, parallel **Python** post-processing on an **88-core** server, and an automated conversion pipeline for vehicle-network data.

Apr 2023 – May 2023 **Advanced AI for Data Analysis (Executive Education)**, *École Polytechnique*, Palaiseau

Sep 2016 – Sep 2018 **M2, Electronics, Electrical & Automation**, *Université Gustave Eiffel*

Sep 2013 – Nov 2016 **Licence, Engineering Science**, *Université Paris-Est Créteil (UPEC)*

Skills

Infrastructure & HPC	**Slurm** (expert), **Singularity** & **Docker**, **uv**; **multi-node GPU** (up to **512 H100s**), NCCL, MPI, Nsight; distributed FFTs; custom CUDA primitives integrated into XLA via FFI/CustomCall.
Programming	**Python** (7y), **JAX** (3y), modern **C++** 17/20 (5y), **CUDA** (3y), **PyTorch** (1y).
ML & Statistics	High-dimensional Bayesian inference (MCMC, HMC, NUTS) with NumPyro / BlackJAX; simulation-based and *field-level inference*; generative deep learning; uncertainty quantification.
Software & DevOps	Clean, modular, well-tested code; **CI/CD** (GitHub / GitLab), TDD, Python packaging (PyPI), **Hugging Face Hub** (Datasets, Spaces), CMake, Linux.
Languages	French (native), Arabic (native), English (C1), German (A2).

Open-Source Software

- `jax-fli` — **Author**. End-to-end differentiable cosmology pipeline in **JAX** (initial conditions → N-body → lensing → power spectra), multi-GPU via sharding, with **Hugging Face Datasets** I/O and a results **Space**.
- `furax-cs` — **Author**. GPU CMB foreground decontamination in **JAX**, built around `ADABK` — a novel minimizer that converges in far fewer iterations than standard optimizers. On **PyPI**, with a live **Hugging Face Space** for the results.
- `FURAX` — **Main contributor & Maintainer**. **JAX** framework for *inverse problems*; the engine behind `furax-cs`.
- `jaxDecomp` — **Author & Maintainer**. **JAX** library for distributed 3D domain decomposition and parallel FFTs: **cuDecomp**/NCCL multi-GPU plus a pure-**JAX** backend portable across GPU / TPU / CPU. *Peer-reviewed in JOSS (2026).*

Selected Publications & Software

Peer-reviewed software

- **Kabalan, W.**, Lanusse, F., Boucaud, A., and Aubourg, E. (2026). jaxDecomp: JAX Library for 3D Domain Decomposition and Parallel FFTs. *Journal of Open Source Software*, 11(121), 8852.

Refereed

- Spagnoletti, A., Boucaud, A., Huertas-Company, M., **Kabalan, W.**, and Biswas, B. (2024). Bayesian Deconvolution of Astronomical Images with Diffusion Models. arXiv:2411.19158 [astro-ph.IM]. *Contribution:* **JAX**-based deconvolution code, multi-node simulations on Jean Zay HPC.
- Sommer, K., **Kabalan, W.**, and Brunet, R. (2024). Infrared Radiometric Image Classification and Segmentation of Cloud Structure Using a Deep-Learning Framework. EGUsphere Preprint 2024-101. *Contribution:* **JAX**-based U-Net implementation and training on Jean Zay HPC.

Preprint

- **Kabalan, W.**, Rizzieri, A., Sohn, W., Basyrov, A., Boucaud, A., Beringue, B., Chanial, P., Tsang King Sang, E., and Errard, J. (2026). A GPU-Accelerated JAX Framework for Robust Parametric Component Separation and Clustering Optimization for CMB Polarization Satellites. arXiv:2604.08463 [astro-ph.CO].

In preparation

- **Kabalan, W.**, Lanusse, F., Boucaud, A., and Aubourg, E. (2026). *Differentiable Forward Modelling for Field-Level Inference in Stage IV Weak Lensing Surveys*.

Collaborations

- **LSST Dark Energy Science Collaboration (DESC)** — Member; *weak-lensing* forward models and scalable, differentiable simulation tools.
- **Simons Observatory** — Member; component-separation tooling (`FURAX`) for *CMB* analysis.

Talks & Tutorials

Oct 2025	**Generative AI with JAX [Tutorial]**	*AISSAI School 2025*
May 2025	**JAXPM: Scalable, Differentiable Particle-Mesh Simulations**	*Bayesian Deep Learning Workshop*
May 2025	**Bayesian Inference for Cosmology with JAX [Tutorial]**	*Bayesian Deep Learning Workshop*
Nov 2024	**Massively Parallel Computing in Cosmology with JAX [Tutorial]**	*CoPhy 2024*
Jun 2024	**Differentiable and Distributed Particle-Mesh N-body Simulations**	*LSST France 2024*

Supervision

Jan–Jun 2025 **Binh Nguyen** — Master 1 student. Project: *Inference of Weak-Lensing Parameters from Blended Galaxies Using Generative Neural Networks* (final report). Outcome: talk at *Rencontres du Vietnam*.